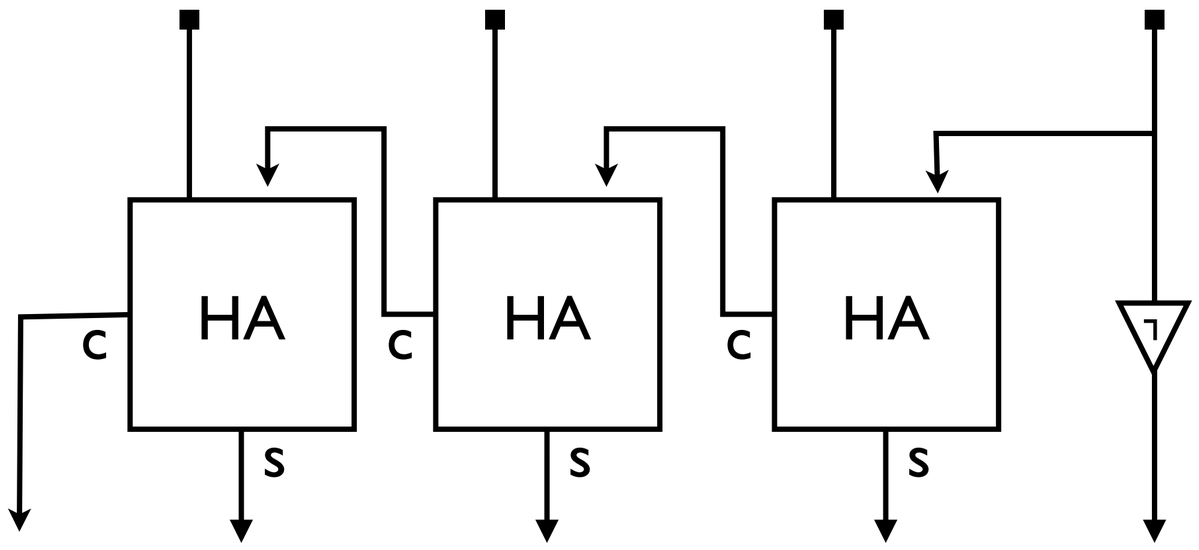
10. Lessons from Binary Successor

The +1 B Combinatorial Circuit



+1[0-15] B combinatorial circuit

Lesson 1: The Physical Realization of Cognition

+1B is a blueprint for the hypothesis that an entirely physical system can perform (basic) rational operations— provided we have a way of establishing representation relations between symbols and contents. A key step in our answer to Leibniz.

> "[T]hree famous philosophical dilemmas stood in the way [of Hobbes's idea]: (i) the metaphysical problem of mind interacting with matter; (ii) the theoretical problem of explaining the relevance of meanings, without appealing to a question-begging homunculus; and (iii) the methodological issue over the empirical testability (and, hence, respectability) of "mentalistic" explanations. The computational idea can be seen as slicing through all three dilemmas at a stroke…" (Haugeland 1981, p. 34)

> "If there are no philosophical dilemmas about chess-playing computers, then why should there be any about chess-playing people – or, indeed, about human intelligence in any other form? To put it coldly: why not suppose that people *just are* computers (and send philosophy packing)?" (Haugeland 1981, p. 36)

Lesson 2: Information Processing as Functional Description

Hammers have the function of pounding nails, and hearts have the function of pumping blood. +1B has an *information processing* function, which goes beyond the way it manipulates the physical world. "M computes +1" is the informational-level description of the machine.

Lesson 3: The Design Impact of Representational Systems

Compare +1B and +1T machines: dramatically different physical functions, same information processing function. Machines and representational systems must be designed together, or co-evolve.

Lesson 4: The Causal Power of Representations

The value of a function depends on the argument: the value of +1 depends on whether you apply it to 2 or to 3, for example. In a computer, these mathematical dependencies are turned into causal dependencies. In the +1 B machine, the *symbol* for 2 *causes* the *symbol* for 3, just the way the number 2 determines the number 3 for the +1 function. A central idea in CTM is that we explain the inferential connections between ideas by the causal connections between symbols.

> "In effect, given an interpreted formal system with true axioms and truth-preserving rules, if you take care of the syntax, *the semantics will take care of itself*."
> (Haugeland 1981, p. 44)

> "An automatic formal system with an interpretation such that the semantics will take care of itself is what Daniel Dennett (1981) calls a *semantic engine*. The discovery that semantic engines are possible – that with the right kind of formal system and interpretation, a machine can handle meanings – is the basic inspiration of cognitive science and artificial intelligence."
> (Haugeland 1981, p. 45)

Lesson 5: Computing a Function vs. Implementing a Function

All things *implement* the laws of physics, but only some things *compute* the laws of physics. To compute is to *represent* a function, but not to actually conform with it. +1B can compute the laws of *adding-apples-to-a-basket*, but it does not *implement* these laws. (By the way, do the planets compute their own orbit? When humans walk, do they compute or implement their trajectory?)

Lesson 6: Computation Does Not Require Explicitly Represented Rules

+1B computes a function not by "following" a set of rules, which would require an inner-agent to do the following, but simply by *embodying* a set of rules in its functional architecture. A machine can compute a function without *understanding* that function in any way; all it has to do is *perform* that function. Compare: the cognitive capacities of insects and babies, but also most adult human cognition, outside of explicit reasoning, including perception, motor control, autonomic functioning, and so on.

> "[The rule-applier] must be regarded as a complete little person – or *homunculus* (in Latin) – inside the head, directing the thoughts like a traffic cop. The trouble is: a theory that invokes an homunculus to *explain* thinking has begged its own question, because the homunculus itself has to think, and *that* thinking has *not* been explained." (Haugeland 1981, p. 35)

Lesson 7: Competence vs. Performance

+1B shows how a physical system can have certain cognitive *capacities*— i.e. the +1B machine has the capacity to compute +1, for a range of arguments, even if it never actually performs this operation. It also illustrates the contrast between the capacities you have *competence* for, and your actual performance. Because of entropy, friction, and the rest, how a machine actually performs can pull apart from its competence. A physical +1B machine won't compute +1 perfectly all the time. In other words, its *performance* may be flawed. We can still say that it is a +1 machine, since this is its underlying *competence*.